H M A Mohit *Chowdhury*

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PhD Researcher in AI/Machine Learning with a solid foundation in deep learning for large-scale time series and multimodal biomedical data, including genomic domain. Strong expertise in the development of AI systems employing self-supervised learning, transformers, and diffusion models for prediction, interpolation, and representation learning on large-scale datasets. Proven expertise in the design of scalable and robust AI systems via the effective use of Python and cloud-based platforms for the deployment of AI systems with a strong research publication portfolio and passion for the translation of research to real-world applications in healthcare domains.

Education

Ph.D. in Computer Science and Engineering	
University of North Texas	*2025 - 2027 (Expected)*
University of Colorado Colorado Springs (Transferred at UNT)	*2022 - 2025*
M.Sc. in Computer Science	**GPA: 3.867**
University of Colorado Colorado Springs	*2024*
B.Sc. in Computer Science and Engineering	**GPA: 3.71**
Ahsanullah University of Science and Technology	*2017*

Scientific Research (Google Scholar)

Journal Article

- **Chowdhury, H. M. A. M.**, Oluwadare, O. *HiCInterpolate: 4D Spatiotemporal Interpolation of Hi-C Data for Genome Architecture Analysis.* **bioRxiv** (2026), https://doi.org/10.64898/2026.02.06.704438
- **Chowdhury, H. M. A. M.**, Oluwadare, O. *EmbedTAD Using Graph Embedding and Unsupervised Learning to Identify TADs from High-Resolution Hi-C Data.* **Communications Biology** 9, 7 (2026), https://doi.org/10.1038/s42003-025-09224-z
- **Chowdhury, H. M. A. M.**, Fuller, M., & Oluwadare, O. *Robin: an advanced tool for comparative loop caller result analysis leveraging large language models.* **NAR Genomics and Bioinformatics** 8(1), 2026, https://doi.org/10.1093/nargab/lqag009
- Pinchuk, D., **Chowdhury, H. M. A. M.**, Pandeya, A., & Oluwadare, O. *HiCForecast: dynamic network optical flow estimation algorithm for spatiotemporal Hi-C data forecasting.* **Bioinformatics** 41(2), 2025, https://doi.org/10.1093/bioinformatics/btaf030
- Rohit, M., **Chowdhury, H. M. A. M.**, & Oluwadare, O. *ScHiCAtt: Hi-C Super Resolution Using Attention Methods for Single Cell.* **Computational and Structural Biotechnology Journal** 27, 2024, https://doi.org/10.1016/j.csbj.2025.02.031
- **Chowdhury, H. M. A. M.**, Boult, T., & Oluwadare, O. *Comparative study on chromatin loop callers using Hi-C data reveals their effectiveness.* **BMC Bioinformatics** 25(1), 2024, https://doi.org/10.1186/s12859-024-05713-w
- Houchens, D., **Chowdhury, H. M. A. M.**, & Oluwadare, O. *coiTAD: Detection of Topologically Associating Domains Based on Clustering of Circular Influence Features from Hi-C Data.* **Genes** 15(10), 2024, https://doi.org/10.3390/genes15101293
- Akpokiro, V., **Chowdhury, H. M. A. M.**, Olowofila, S., Nusrat, R., & Oluwadare, O. *CNNSplice: Robust models for splice site prediction using convolutional neural networks.* **Computational and Structural Biotechnology Journal** 21, 2023, https://doi.org/10.1016/j.csbj.2023.05.031

Poster Presentation

- EmbedTAD: Using Graph Embedding and Unsupervised Learning to Identify TADs from High-Resolution Hi-C Data at Colorado AI Summit, 2025.
- Comparative study on chromatin loop callers using Hi-C data reveals their effectiveness at RECOMB 2024 and UCCS MLRD 2023.

Professional Service

- Reviewer, *Scientific Reports* (Nature Portfolio)
- Reviewer, *BioData Mining* (BMC)
- Reviewer, IEEE International Conference on Bioinformatics and Biomedicine (BIBM)
- Mentor, UR2PhD Program, University of North Texas (Fall 2025 – Spring 2026).

Skills

- **AI / ML**: Deep Learning, Self-Supervised Learning, Transformer Architectures, CNNs, Graph Representation Learning, Clustering, Time-Series Modeling, Spatiotemporal Modeling, Multimodal Learning
- **Programming Languages**: Python, Java, C#, C, JavaScript
- **ML & Data Framework**: PyTorch, RAPIDS (cuML, cuDF), scikit-learn
- **Software Framework**: Spring, .NET, Flask
- **Cloud & Deployment**: AWS (EC2, S3, etc), Docker, Terraform, Git
- **Databases**: MySQL, MSSQL, PostgreSQL
- **Operating Systems**: Linux, macOS, Windows

Experience

Graduate Research Assistant **2022 - Present**
Oluwadare Lab

- Developed advanced AI and ML algorithms for large scale, high-dimensional biological **time-series** data, with a focus on **spatiotemporal modeling** of 3D genome organization using Hi-C data.
- Developed deep learning models for **time series forecasting, interpolation** of Hi-C contact maps to predict missing and future interaction states based on the available data.
- Developed **computer vision**-inspired architectures for modeling **spatiotemporal** contact matrices, aiding the subsequent analysis of chromatin dynamics.
- Proposed and implemented a **graph-based unsupervised** learning framework on Topologically Associating Domain (TAD) detection based on **network embeddings** and **clustering** techniques.
- Developing and training **CNN**-based deep learning models for the detection of biological signals, including the prediction of splice sites from genomic data.
- Performed large-scale **comparative benchmarking** of chromatin loop detection algorithms in terms of robustness, accuracy, and scalability.
- Created and delivered **web-based AI** tools for genomic data analysis utilizing **Python, Flask, Docker, and Cloud Technology**, which improved accessibility to such tools for domain experts.
- Worked in a collaborative research environment where contributions were made to data engineering, model validation, experimental design, and maintenance.
- Co-authored a number of peer-reviewed publications and was a manuscript reviewer for the top journals and conferences in AI and computational biology.

Solution Developer **2022 – 2022**
Otto International

- Designed and implemented a highly scalable **AWS**-based infrastructure for ESB using **Terraform**, thus allowing for automated and reproducible deployments.
- Perform performance and reliability enhancements by refactoring core services with the use of **Java and PostgreSQL**.
- Implement and maintain CI/CD pipelines using Jenkins to make the release process more stable and guarantee efficient deployment.
- Worked with business analysts and engineering teams to solidify requirements and integrate enterprise applications through the ESB.
- Provided technical documentation and supported the knowledge transfer to guarantee scalability and maintainability of the system for a longer period.

Software Engineer **2020 – 2022**
Dovetail Technology

- Design and development of an efficient system using **C#, .NET Web API, and MSSQL** to support core business operations.

- Deployed and maintained the system on **IIS** for live production usage.
- Developed Windows services utilizing a microservices architecture design, leading to improved system reliability and fault tolerance.
- Worked with the front end teams to assist in integration with **AngularJS**.
- Worked closely with BA and automation teams to streamline the process and development flow.

Software Engineer **2019 – 2020**
IdeaScale Bangladesh

- Utilized **Java, Spring, MySQL, and Docker** to enrich backend services for accommodating dynamic business needs.
- Improved platform stability and performance through systematic bug fixes and refactoring.
- Refactored old SQL queries in **JPA and Querydsl**, enhancing code maintainability and programmer productivity.
- Collaborated with cross-functional groups in support of scalable system evolution and process improvements.

Software Engineer **2017 – 2019**
Netweaver Software Limited

- Developed two e-commerce platforms using **Java, Spring Boot, MySQL, SAPUI5, and JavaScript**, and successfully designed and implemented both
- Created a directory-based platform in the realms of **healthcare** focused on doctors and hospitals, utilizing **Node.js, Vue.js, and PostgreSQL**.
- Focused on responsive design, backend scalability, and seamless frontend-backend integration.

Awards

- CENG Anticipated Award, 2025 - University of North Texas.
- Graduate School Travel Award, 2024 - University of Colorado Colorado Springs.
- Tuition Matching Grant Award, 2023-2024 - University of Colorado Colorado Springs.
- Tuition Matching Grant Award, 2022-2023 - University of Colorado Colorado Springs.

Certificates

- Tableau Fundamentals.
- Responsible Conduct of Research (RCR-mini).

Activities

- Participant, Open House 2025 at UCCS.
- President, Bangladeshi Student Association (BSA) at UCCS (2024-2025).
- Participant, Cool Science Festival 2024 at UCCS.
- Session Chair, NSF REU Summer Research 2023 at UCCS.
- Technical Assistant, NSF REU Summer Research 2023 at UCCS.
- Participant, Cool Science Festival 2023 at UCCS.
- Conducted a basic Linux and Docker hands-on class in NSF REU Summer Research 2023 at UCCS.
- Participant, Cool Science Festival 2022 at UCCS.
- Organized *Cumilla Victoria Govt. College Reunion 2016*.